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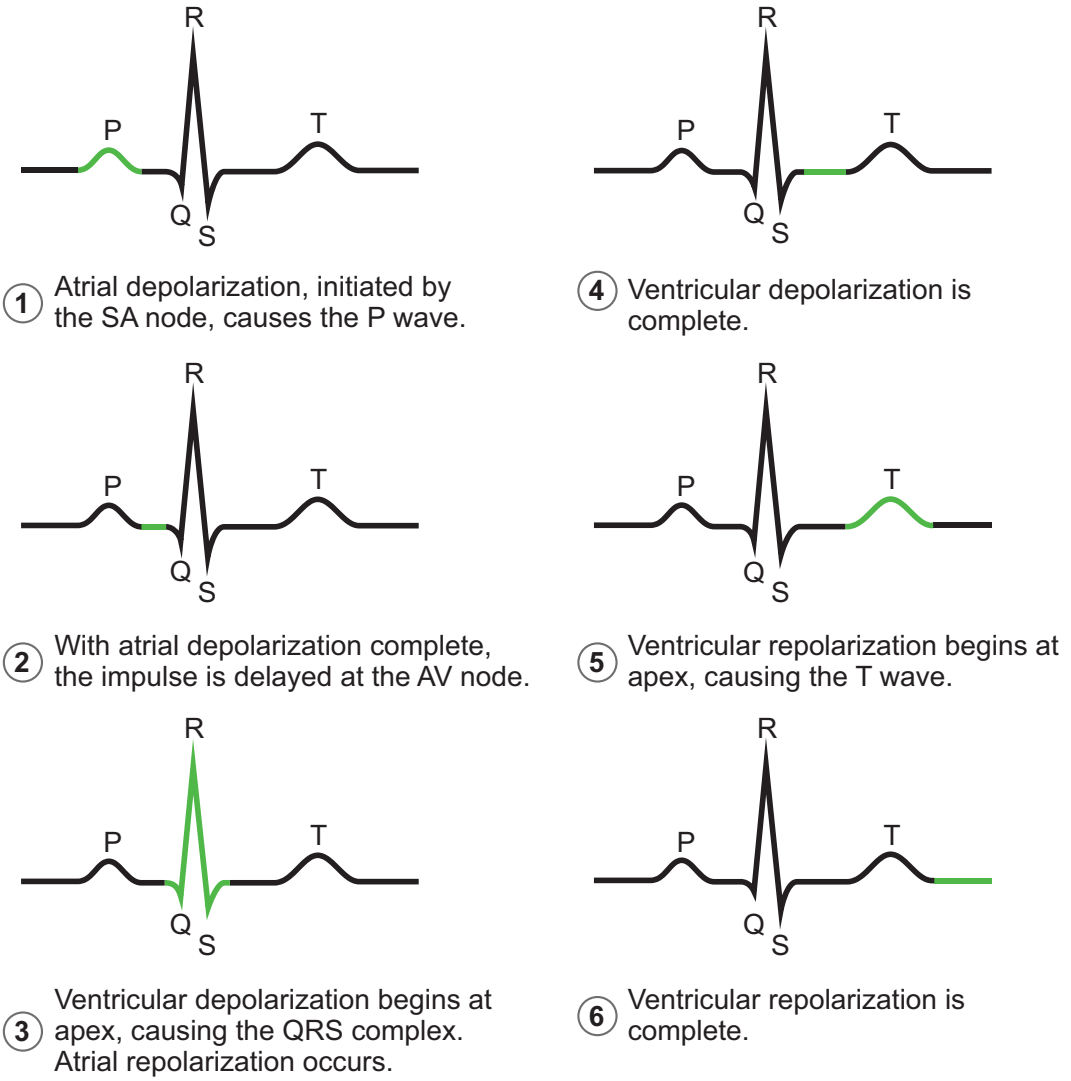

Figure 7.10 Stages of ECG graph

7.5. Double circulation

Circulation of the blood was first described by William Harvey (1628). There are two types of blood circulation in vertebrates, single circulation and double circulation which is shown in Figure 7.11 (a and b) and 7.12.

The blood circulates twice through the heart first on the right side then on the left side to complete one cardiac cycle. The complete double blood circulation is more prominent in mammals because of the complete partition of all the chambers (Auricles and ventricles) in the heart.

In systemic circulation, the oxygenated blood entering the aorta from the left ventricle is carried by a network of arteries, arterioles and capillaries to the tissues. The deoxygenated blood from the tissue is collected by venules, veins and vena cava and emptied into the right atrium. In pulmonary circulation, the blood from heart (right ventricle) is taken to the lungs by pulmonary artery and the oxygenated blood from the lungs is emptied into the left auricle by the pulmonary vein.

Completely separated circuits have an important advantage. Different pressures are maintained in the pulmonary